Chapter 10. The Lebesgue Integral

Lebesgue measure, measurable functions, convergence theorems, L1 and L2, Riesz–Fischer

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10.1 Measurable sets and Lebesgue measure

Motivation. The Riemann integral handles poorly functions with many discontinuities and limits of sequences of functions. Lebesgue's idea is to partition the range instead of the domain, which requires measuring the size of arbitrary sets of the form $\{x : f(x) > c\}$. Lebesgue measure extends length to such sets.

Proposition 10.1 (10.1.1)

Every open $U \subset \mathbb{R}$ is a union of countably many disjoint open intervals, and this representation is unique.

Proof. For $x \in U$ let $I_x = (a_x, b_x)$ with $a_x = \inf\{a : (a, x] \subset U\}$, $b_x = \sup\{b : [x, b) \subset U\}$; $I_x \subset U$ is the largest open interval in U containing x. If $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval in U, so $I_x = I_y$. Each I_x contains a rational, and distinct ones are disjoint, so there are countably many. The intervals are the components of U, which gives uniqueness. ■

For $U = \bigsqcup(a_n, b_n)$ put $\lambda(U) = \sum(b_n - a_n)$ (independent of the order, Proposition 2.16 of Volume I), and for any $S \subset \mathbb{R}$ define the **outer measure**

$$\mu(S) = \inf\{\lambda(U) : U \supset S \text{ open}\} \in [0, \infty].$$

Proposition 10.2 (10.1.2)

(a) $S \subset T \Rightarrow \mu(S) \leq \mu(T)$.
(b) If I is an interval with end points $a \leq b$, then $\mu(I) = b - a$.
(c) $\mu(x + S) = \mu(S)$.
(d) $\mu(\bigcup S_n) \leq \sum \mu(S_n)$ for countably many S_n.

Proof. First, if an open interval (a, b) is covered by open intervals J_k, then $b - a \leq \sum |J_k|$: for $a < c < d < b$ finitely many J_k cover the compact $[c, d]$, and by choosing a chain of overlapping intervals from c to d, $d - c \leq \sum |J_k|$; let $c \to a$, $d \to b$. Summing over components (Proposition 6.9), $\lambda(U) \leq \sum \lambda(U_n)$ whenever $U \subset \bigcup U_n$ are open.

(a) is immediate.

(b) For $I = [a, b]$, $\mu(I) \leq \lambda((a - \epsilon, b + \epsilon)) = b - a + 2\epsilon$; and an open $U \supset I$ has a component containing $[a, b]$, so $\lambda(U) \geq b - a$. Other intervals are similar.

(c) Translation preserves λ of open sets.

(d) We may assume all $\mu(S_n) < \infty$. Choose open $U_n \supset S_n$ with $\lambda(U_n) < \mu(S_n) + \epsilon 2^{-n}$; then $\mu(\bigcup S_n) \leq \lambda(\bigcup U_n) \leq \sum \lambda(U_n) < \sum \mu(S_n) + \epsilon$. ■

In particular countable sets have outer measure 0, so $[0, 1]$ is uncountable. A set $E \subset \mathbb{R}$ is **measurable** if

$$\mu(X) = \mu(X \cap E) + \mu(X \cap E^c) \qquad \text{for every } X \subset \mathbb{R};$$

let $\mathfrak{M}$ be the family of measurable sets. By (d), only $\geq$ needs checking, and only for $\mu(X) < \infty$.

Theorem 10.3 (10.1.3)

(a) $\emptyset \in \mathfrak{M}$.
(b) $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$.
(c) $E, F \in \mathfrak{M} \Rightarrow E \cup F \in \mathfrak{M}$.
(d) $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$.

(e) If $\mu(E) = 0$, then $E \in \mathfrak{M}$.
(f) Every interval is measurable.
(g) Every open set is measurable.

Proof. (a), (b) are clear from the symmetric definition.

(c) Since $X \cap (E \cup F) = (X \cap E) \cup (X \cap F \cap E^c)$,

$$\mu(X\cap(E\cup F))+\mu(X\cap(E\cup F)^c) \le \mu(X\cap E)+\mu(X\cap E^c\cap F)+\mu(X\cap E^c\cap F^c) = \mu(X\cap E)+\mu(X\cap E^c) = \mu(X),$$

using $F \in \mathfrak{M}$ (for the set $X \cap E^c$) and then $E \in \mathfrak{M}$.

(d) Let $F_n = E_n - (E_1 \cup \cdots \cup E_{n-1})$ (measurable by (b), (c), as $A \cap B = (A^c \cup B^c)^c$), disjoint, and $G_n = F_1 \cup \cdots \cup F_n$, $E = \bigcup E_n = \bigsqcup F_n$. Using $F_n \in \mathfrak{M}$ for the set $X \cap G_n$, $\mu(X \cap G_n) = \mu(X \cap G_{n-1}) + \mu(X \cap F_n)$, so $\mu(X \cap G_n) = \sum_{i\le n} \mu(X \cap F_i)$. As $G_n \in \mathfrak{M}$ and $G_n^c \supset E^c$,

$$\mu(X) = \mu(X \cap G_n) + \mu(X \cap G_n^c) \ge \sum_{i\le n} \mu(X \cap F_i) + \mu(X \cap E^c).$$

Let $n \to \infty$ and use (d) of Proposition 10.2: $\mu(X) \ge \mu(X \cap E) + \mu(X \cap E^c)$.

(e) $\mu(X \cap E) = 0$ and $\mu(X \cap E^c) \le \mu(X)$.

(f) By (b), (d) and (e) it suffices to treat $I = (a, \infty)$. Given X and ϵ, take open $U \supset X$ with $\lambda(U) < \mu(X)+\epsilon$. The open sets $U\cap(a,\infty)$, $U\cap(-\infty,a)$ satisfy $\lambda(U\cap(a,\infty))+\lambda(U\cap(-\infty,a)) \le \lambda(U)$ (each component of U is split into two intervals), so $\mu(X \cap I) + \mu(X \cap I^c) \le \lambda(U \cap (a,\infty)) + \lambda(U \cap (-\infty, a)) + \mu(\{a\}) < \mu(X) + \epsilon$.

(g) Proposition 10.1, (f) and (d). ■

So $\mathfrak{M}$ is a σ-algebra containing all Borel sets. The **Lebesgue measure** is $m = \mu|_{\mathfrak{M}}$.

Theorem 10.4 (10.1.4, countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigsqcup E_n) = \sum m(E_n)$.

Proof. In the proof of Theorem 10.3(d) take $X = E = \bigsqcup E_n$ (so $F_n = E_n$): $m(E) \ge \sum m(E_n)$; the reverse is Proposition 10.2(d). ■

Consequences: if $E_1 \subset E_2 \subset \cdots$, then $m(\bigcup E_n) = \lim m(E_n)$ (write the union as $\bigsqcup(E_n - E_{n-1})$); if $E_1 \supset E_2 \supset \cdots$ and $m(E_1) < \infty$, then $m(\bigcap E_n) = \lim m(E_n)$. By definition of μ, every measurable E satisfies $m(E) = \inf\{m(U) : U \supset E \text{ open}\}$, and for bounded E, $m(E) = \sup\{m(K) : K \subset E \text{ compact}\}$ (apply the outer approximation to $[a,b] - E$). Not every set is measurable (Vitali): choosing one point in $[0,1)$ from each coset of $\mathbb{Q}$ gives a set V whose rational translates mod 1 are disjoint and cover $[0,1)$, so $m(V)$ could be neither 0 nor positive.

10.2 Measurable functions and the integral

Motivation. With measure in hand, we integrate simple functions and then approximate. Measurable functions are closed under all limit operations, so this class is large enough for analysis.

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$. $f : E \to \mathbb{R}^*$ is **measurable** if $\{x \in E : f(x) > a\} \in \mathfrak{M}$ for every $a \in \mathbb{R}$; a complex function is measurable if its real and imaginary parts are.

Proposition 10.5 (10.2.1)

For $f : E \to \mathbb{R}^*$ the following are equivalent:
(a) f is measurable;
(b) $\{f \geq a\} \in \mathfrak{M}$ for all a;
(c) $\{f < a\} \in \mathfrak{M}$ for all a;
(d) $\{f \leq a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \geq a\} = \bigcap_n \{f > a - 1/n\}$, $\{f > a\} = \bigcup_n \{f \geq a + 1/n\}$, and (c), (d) are the complements (relative to E) of (b), (a). ∎

Corollary 10.6 (10.2.2)

If f is measurable, then $\{f = a\}$ is measurable for every $a \in \mathbb{R}^*$.

Proof. $\{f = a\} = \{f \geq a\} \cap \{f \leq a\}$ for real a, and $\{f = \infty\} = \bigcap_n \{f \geq n\}$. ∎

Proposition 10.7 (10.2.3)

(a) Continuous $f : E \to \mathbb{R}$ are measurable.
(b) If f, g are measurable and real-valued and $a \in \mathbb{R}$, then $f + g$, fg, af are measurable.
(c) If each f_n is measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf f_n$, $\limsup f_n$, and hence $\lim f_n$ where it exists.

Proof. (a) $\{f > a\} = U \cap E$ with U open.

(b) $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > r\} \cap \{g > a - r\}$; f^2 is measurable since $\{f^2 > a\} = \{f > \sqrt{a}\} \cup \{f < -\sqrt{a}\}$ for $a \geq 0$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

(c) $\{\sup f_n > a\} = \bigcup \{f_n > a\}$, $\inf f_n = -\sup(-f_n)$, $\limsup f_n = \inf_n \sup_{k \geq n} f_k$. ∎

A **simple function** is $s = \sum_{i=1}^k a_i \chi_{A_i}$ with $a_i \geq 0$ and $A_i \in \mathfrak{M}$; its integral is $\int_E s = \sum a_i m(A_i \cap E)$ (independent of the representation, by passing to a common disjoint refinement and using additivity). For measurable $f : E \to [0, \infty]$,

$$\int_E f = \sup\Big\{ \int_E s : 0 \leq s \leq f,\ s \text{ simple} \Big\}.$$

For simple functions the two definitions agree, and $0 \leq f \leq g$ implies $\int f \leq \int g$.

Proposition 10.8 (10.2.4)

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions.

Proof. Let $A_{n,i} = \{\frac{i-1}{2^n} \leq f < \frac{i}{2^n}\}$ $(1 \leq i \leq n2^n)$, $B_n = \{f \geq n\}$, and $s_n = \sum_i \frac{i-1}{2^n} \chi_{A_{n,i}} + n\chi_{B_n}$. Then $s_n \leq s_{n+1} \leq f$, and $f - s_n < 2^{-n}$ where $f < n$; also $s_n = n \to \infty$ where $f = \infty$. ∎

Theorem 10.9 (10.2.5)

If $A = \bigsqcup A_n$ with $A_n \in \mathfrak{M}$ disjoint and $f : A \to [0, \infty]$ is measurable, then $\int_A f = \sum_n \int_{A_n} f$.

Proof. For simple s, $\int_A s = \sum_n \int_{A_n} s$ by Theorem 10.4; so for simple $s \leq f$, $\int_A s \leq \sum \int_{A_n} f$, and taking the supremum, $\int_A f \leq \sum \int_{A_n} f$. Conversely, given N and ϵ, choose simple $s_k \leq f$ on A_k with $\int_{A_k} s_k > \int_{A_k} f - \epsilon/N$ $(k \leq N)$; $s = \sum_{k \leq N} s_k \chi_{A_k} \leq f$, so $\int_A f \geq \sum_{k \leq N} \int_{A_k} f - \epsilon$. ∎

So $A \mapsto \int_A f$ is a measure. For measurable f, write $f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$; $f \in L^1(E)$ if $\int_E |f| < \infty$, and then $\int_E f = \int_E f^+ - \int_E f^-$; for complex f, $\int f = \int \operatorname{Re} f + i \int \operatorname{Im} f$.

Corollary 10.10 (10.2.6)

Theorem 10.9 holds for every $f \in L^1(A)$, the series converging absolutely.

Proof. Apply Theorem 10.9 to $f^{\pm}$ (real and imaginary parts) and subtract; the series converge because $\sum \int_{A_n} |f| = \int_A |f| < \infty$. ■

Corollary 10.11 (10.2.7)

A measurable f is in $L^1(E)$ iff $|f| \in L^1(E)$, and then $|\int_E f| \le \int_E |f|$.

Proof. The first statement is the definition. Choose θ with $e^{-i\theta} \int f = |\int f|$; then $|\int f| = \int \operatorname{Re}(e^{-i\theta} f) \le \int |f|$ (using linearity, Theorem 10.14). ■

Proposition 10.12 (10.2.8)

If $f \ge 0$ is measurable and $\int_E f = 0$, then $f = 0$ almost everywhere.

Proof. With $A_n = \{f \ge 1/n\}$, $\frac{1}{n} m(A_n) \le \int f = 0$, so $m(\{f > 0\}) = m(\bigcup A_n) = 0$. ■

10.3 Convergence theorems

Motivation. The payoff of the Lebesgue integral: limits and integrals can be exchanged under very mild conditions (monotone convergence, Fatou, dominated convergence), with no uniform convergence required.

Theorem 10.13 (10.3.1, monotone convergence)

If $0 \le f_1 \le f_2 \le \cdots$ are measurable on E and $f = \lim f_n$, then $\int_E f = \lim \int_E f_n$.

Proof. f is measurable and $\int f_n \le \int f_{n+1} \le \int f$, so $L = \lim \int f_n \le \int f$. Let $0 \le s \le f$ be simple and $0 < c < 1$; $E_n = \{f_n \ge cs\}$ increase to E (where $f(x) = 0$, $x \in E_1$; otherwise $cs(x) < f(x)$). Then $\int f_n \ge \int_{E_n} f_n \ge c \int_{E_n} s \to c \int_E s$ (Theorem 10.9, continuity from below). So $L \ge c \int s$; let $c \to 1$ and take the supremum over s. ■

Theorem 10.14 (10.3.2, linearity)

(a) If $f, g \in L^1(E)$, then $f + g \in L^1(E)$; if $f, g \ge 0$ or $f, g \in L^1(E)$, then $\int (f + g) = \int f + \int g$.
(b) If $f \in L^1(E)$ and $a \in \mathbb{C}$, then $af \in L^1(E)$ and $\int af = a \int f$.

Proof. For simple functions this is immediate. For $f, g \ge 0$ take increasing simple $s_n \to f$, $t_n \to g$ (Proposition 10.8); $s_n + t_n \uparrow f + g$, and Theorem 10.13 gives the formula. Then $|f + g| \le |f| + |g|$ gives integrability. For real $f, g \in L^1$ and $h = f + g$, $h^+ + f^- + g^- = h^- + f^+ + g^+$; integrate and rearrange. Complex functions: real and imaginary parts.

(b) is similar (for $a = i$ the parts are exchanged). ■

Corollary 10.15 (10.3.3)

If $f_n : E \to [0,\infty]$ are measurable, then $\int_E \sum_n f_n = \sum_n \int_E f_n$.

Proof. Apply Theorem 10.13 to the partial sums, using Theorem 10.14. ■

Theorem 10.16 (10.3.4, Riemann and Lebesgue integrals)

Let $f : [a,b] \to \mathbb{R}$ be bounded.
(a) If f is Riemann integrable, then f is measurable and its Lebesgue integral equals its Riemann integral.
(b) f is Riemann integrable iff the set of its discontinuities has measure 0.

Proof. Choose partitions $P_1 \subset P_2 \subset \cdots$ with $\|P_k\| \to 0$, and let ℓ_k, u_k be the step functions equal to m_i, M_i on the open subintervals of P_k (and to f at partition points). Then $\ell_k \uparrow \ell$, $u_k \downarrow u$, $\ell \le f \le u$, and by monotone convergence (applied to $u_1 - u_k$ and $\ell_k - \ell_1$), $\int \ell = \lim L(f,P_k)$ and $\int u = \lim U(f,P_k)$. Let $P = \bigcup P_k$, a countable set. For $x \notin P$, $u(x)$ and $\ell(x)$ are the limits of sup and inf of f over shrinking neighbourhoods of x, so f is continuous at x iff $u(x) = \ell(x)$.

If f is Riemann integrable, then $U(f,P_k) - L(f,P_k) \to 0$ (proof of Theorem 5.8 of Volume I), so $\int (u-\ell) = 0$ and $u = \ell$ a.e. (Proposition 10.12). Then $f = u$ a.e.; f is measurable (Theorem 10.3(e)) and $\int f = \int u = \lim U(f,P_k)$ is the Riemann integral. This proves (a) and, with the characterization of continuity, the necessity in (b). Conversely, if the discontinuities have measure 0, then $u = \ell$ a.e., so $\lim(U(f,P_k) - L(f,P_k)) = \int (u-\ell) = 0$, and f is Riemann integrable by Riemann's criterion. ■

Theorem 10.17 (10.3.5, Fatou's lemma)

If $f_n : E \to [0,\infty]$ are measurable, then $\int_E \liminf f_n \le \liminf \int_E f_n$.

Proof. $g_n = \inf_{k \ge n} f_k$ increases to $\liminf f_n$ and $\int g_n \le \int f_k$ for $k \ge n$; apply Theorem 10.13. ■

Theorem 10.18 (10.3.6, Lebesgue's dominated convergence theorem)

Let $f_n : E \to \mathbb{C}$ be measurable with (a) $|f_n| \le g$ for some $g \in L^1(E)$ and (b) $f(x) = \lim f_n(x)$ for each x. Then $f \in L^1(E)$ and $\lim \int_E |f_n - f| = 0$; in particular $\int f_n \to \int f$.

Proof. $|f| \le g$, so $f \in L^1$. Apply Fatou to $2g - |f_n - f| \ge 0$: $\int 2g \le \liminf \int (2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. Hence $\limsup \int |f_n - f| \le 0$. ■

The theorems remain valid when the hypotheses hold only almost everywhere, since sets of measure 0 do not affect integrals.

Theorem 10.19 (10.3.7, Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int_E |f_n| < \infty$, then $\sum f_n(x)$ converges absolutely for almost every x, its sum f is in $L^1(E)$, and $\int_E f = \sum \int_E f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ (Corollary 10.15), so $g < \infty$ a.e. There the series converges absolutely, with partial sums dominated by g; apply Theorem 10.18. ■

Theorem 10.20 (10.3.8, absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ for all measurable $A \subset E$ with $m(A) < \delta$.

Proof. $\min(|f|, n) \uparrow |f|$, so by monotone convergence choose N with $\int(|f| - \min(|f|, N)) < \epsilon/2$, and $\delta = \epsilon/(2N)$. Then $\int_A |f| \le \epsilon/2 + Nm(A) < \epsilon$. ∎

10.4 The spaces L^1 and L^2

Motivation. With the Lebesgue integral, L^1 and L^2 become complete. L^2 is a Hilbert space in which Fourier series converge in norm and Parseval's identity holds; this is the clean form of the Fourier theory of Chapter 9.

Identifying functions that agree a.e., $L^1(E)$ with $\|f\|_1 = \int |f|$ and $L^2(E)$ (measurable f with $\int |f|^2 < \infty$) with $\|f\|_2 = (\int |f|^2)^{1/2}$ are normed spaces; L^2 has the inner product $\langle f, g\rangle = \int f\bar{g}$ ($|f\bar{g}| \le \frac{1}{2}(|f|^2 + |g|^2)$).

Theorem 10.21 (10.4.1)

Every Cauchy sequence in $L^1(E)$ converges.

Proof. By Proposition 7.11 it suffices that $\sum \|f_n\|_1 < \infty$ implies convergence of $\sum f_n$ in L^1. By Theorem 10.19, $f = \sum f_n$ exists a.e. and is in L^1, and $\|f - \sum_{n \le N} f_n\|_1 \le \sum_{n>N} \|f_n\|_1 \to 0$. ∎

Theorem 10.22 (10.4.2)

$L^2(E)$ is a Banach space (a Hilbert space).

Proof. Let $\sum \|f_n\|_2 = M < \infty$ and $g = \sum |f_n|$. By the triangle inequality $\|\sum_{n \le N} |f_n|\|_2 \le M$, and monotone convergence gives $\int g^2 \le M^2$, so $g < \infty$ a.e. and $f = \sum f_n$ exists a.e. with $|f| \le g$. Since $|f - \sum_{n \le N} f_n|^2 \le g^2$ and tends to 0 a.e., dominated convergence gives $\|f - \sum_{n \le N} f_n\|_2 \to 0$. Apply Proposition 7.11. ∎

Theorem 10.23 (10.4.3, Riesz–Fischer)

For every $c \in \ell^2(\mathbb{Z})$ there is a periodic $f \in L^2[-\pi, \pi]$ with $\hat{f}(n) = c(n)$ for all n.

Proof. By orthonormality, $s_N = \sum_{|n| \le N} c(n) e^{inx}$ satisfies $\frac{1}{2\pi}\int |s_M - s_N|^2 = \sum_{N < |n| \le M} |c(n)|^2$, so (s_N) is Cauchy in L^2 and converges to some f (Theorem 10.22). By Cauchy–Schwarz, $\hat{f}(n) = \lim_N \widehat{s_N}(n) = c(n)$. ∎

Thus $f \mapsto \hat{f}$ is an isometric isomorphism of $L^2[-\pi, \pi]$ (with $\frac{1}{2\pi}\int$) onto $\ell^2(\mathbb{Z})$, once injectivity is known (Theorem 10.26).

Proposition 10.24 (10.4.4)

If $f \in L^1(E)$ and $\int_A f = 0$ for every measurable $A \subset E$, then $f = 0$ a.e.

Proof. With $A = \{\operatorname{Re} f > 0\}$, $\int_A \operatorname{Re} f = \operatorname{Re} \int_A f = 0$, so $\operatorname{Re} f \chi_A = 0$ a.e. (Proposition 10.12) and $m(A) = 0$. Similarly for $\{\operatorname{Re} f < 0\}$, $\{\operatorname{Im} f > 0\}$, $\{\operatorname{Im} f < 0\}$. ∎

Lemma 10.25 (10.4.5)

For every measurable $E \subset [a,b]$ there are continuous g_n with $|g_n| \le 1$ and $g_n \to \chi_E$ a.e.

Proof. By regularity choose compact K_n and open U_n with $K_n \subset E \subset U_n$ and $m(U_n - K_n) < 2^{-n}$, and let $g_n(x) = \frac{d(x,U_n^c)}{d(x,U_n^c)+d(x,K_n)}$, continuous, equal to 1 on K_n and 0 off U_n. Then $g_n(x) \ne \chi_E(x)$ only for $x \in U_n - K_n$. The set of x lying in infinitely many $U_n - K_n$ is contained in $\bigcup_{n \ge N}(U_n - K_n)$ for every N, so has measure at most 2^{1-N}, i.e. 0. Elsewhere $g_n(x) \to \chi_E(x)$. ■

Theorem 10.26 (10.4.6)

If $f \in L^1[-\pi,\pi]$ and $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ a.e.

Proof. By hypothesis $\int fg = 0$ for every trigonometric polynomial g, hence for every continuous periodic g (Fejér, Theorem 9.11, gives uniformly convergent trigonometric polynomials; pass to the limit by dominated convergence). Let $A \subset [-\pi+\eta, \pi-\eta]$ be measurable. Lemma 10.25, with the U_n inside $(-\pi,\pi)$, gives continuous periodic g_n, $|g_n| \le 1$, $g_n \to \chi_A$ a.e.; by dominated convergence $\int_A f = \lim \int f g_n = 0$. Letting $\eta \to 0$, $\int_A f = 0$ for all measurable A, and Proposition 10.24 applies. ■

Hence two L^1 functions with the same Fourier coefficients agree a.e., and Parseval's identity holds for every $f \in L^2[-\pi,\pi]$.

Lemma 10.27 (10.4.7)

For $f \in L^2(E)$ and $\epsilon > 0$ there is a simple function s (a finite complex combination of characteristic functions of sets of finite measure) with $\|f - s\|_2 < \epsilon$.

Proof. For $f \ge 0$ take increasing simple $s_n \to f$ (Proposition 10.8); $|f - s_n|^2 \le f^2$, so $\|f - s_n\|_2 \to 0$ by dominated convergence, and $\int s_n^2 < \infty$ forces the sets in s_n to have finite measure. In general use $f = (\operatorname{Re} f)^+ - (\operatorname{Re} f)^- + i(\operatorname{Im} f)^+ - i(\operatorname{Im} f)^-$. ■

Theorem 10.28 (10.4.8)

For $f \in L^2[a,b]$ and $\epsilon > 0$ there is $g \in C[a,b]$ with $\|f - g\|_2 < \epsilon$.

Proof. By Lemma 10.27 it suffices to approximate χ_A, $A \subset [a,b]$ measurable. With g_n from Lemma 10.25, $|g_n - \chi_A|^2 \le 4$ and tends to 0 a.e., so $\|g_n - \chi_A\|_2 \to 0$ by dominated convergence on the finite interval. ■

Theorem 10.29 (10.4.9)

For $f \in L^1(E)$ and $\epsilon > 0$ there is a continuous g on $\mathbb{R}$ vanishing outside a bounded interval with $\|f - g\|_1 < \epsilon$ (norm taken over E).

Proof. By dominated convergence $\|f - f\chi_{[-N,N]}\|_1 \to 0$; approximate $f\chi_{[-N,N]}$ in L^1 by simple functions as in Lemma 10.27; approximate each $\chi_{A \cap E}$, $A \subset [-N,N]$, by g_n from Lemma 10.25 (with $U_n \subset (-N-1, N+1)$), using dominated convergence. ■

In particular L^1 and L^2 are the completions of the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ of Chapter 7.

10.5 The Poisson summation formula

Motivation. The Fourier transform extends Fourier series to nonperiodic functions on $\mathbb{R}$. The Poisson summation formula connects the two and has applications from theta functions to sampling theory.

For $g \in L^1(\mathbb{R})$ let $G(x) = \sum_{n\in\mathbb{Z}} g(x + 2\pi n)$.

Theorem 10.30 (10.5.1)

The series defining G converges absolutely for almost every x, $G \in L^1[-\pi, \pi]$, and

$$\hat{G}(n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

Proof. Integrals over $\mathbb{R}$ are translation invariant (Proposition 10.2(c) for sets, hence for simple functions and by monotone convergence for all $f \geq 0$). By Corollary 10.15, $\int_{-\pi}^{\pi}\sum_n |g(x+2\pi n)|\,dx = \sum_n \int_{-\pi+2\pi n}^{\pi+2\pi n}|g| = \int_{\mathbb{R}}|g| < \infty$. So the series converges absolutely a.e. and $G \in L^1$. By Theorem 10.19,

$$\hat{G}(n) = \frac{1}{2\pi}\sum_k \int_{-\pi}^{\pi} g(x + 2\pi k)e^{-inx}dx = \frac{1}{2\pi}\sum_k \int_{-\pi+2\pi k}^{\pi+2\pi k} g(y)e^{-iny}dy = \frac{1}{2\pi}\int_{\mathbb{R}} ge^{-iny}dy.$$

■

If moreover the Fourier series of G converges to $G(0)$ at 0 (for example if g is smooth and rapidly decreasing), this gives the **Poisson summation formula** $\sum_n g(2\pi n) = \frac{1}{2\pi}\sum_n \int_{\mathbb{R}} g(x)e^{-inx}dx$. Applied to $g(x) = e^{-ax^2}$ it yields the transformation formula for the theta function $\theta(x) = \sum_n e^{-\pi n^2 x}$: $\theta(1/x) = \sqrt{x}\,\theta(x)$.